Thushar Reddy Vemula

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Experience

Software Automation Aide Jul 2026 – Present
Arizona State University – Enterprise Brand Strategy & Management Tempe, AZ

- Automated bug intake, routing, and status synchronization through Python and Airtable/Jira REST APIs, cutting triage and handoff time by **65%** and saving **8 engineering hours weekly**.
- Established Python/Playwright regression coverage for DOM structure, metadata, responsive layouts, and WCAG checks, maintaining **92%+ quality and accessibility scores** across **8 production websites**.
- Prioritized defects by GA4 pageview volume and severity, accelerating resolution turnaround on high-traffic pages by **30%**.

Software Engineer – Founding Project Dec 2025 – May 2026
SimCoach – Live AI Rehearsal Platform

- Shipped a Next.js/Supabase job application platform serving **1,000+ active users**, alongside a Chrome extension that detects and parses job listings on LinkedIn and Indeed.
- Engineered an AI mock interview engine combining intent-based model routing with LiveKit audio/video, sustaining **sub-300 ms media latency** across **1,000+ sessions**.
- Reduced LLM API costs by **90%** and achieved **2-second p95 response latency** through Redis/ElastiCache semantic-context caching and Llama 3.3 70B fine-tuning with SFT and LoRA on GCP Model Garden.
- Implemented a multi-tenant workspace propagating application records, ownership, and status changes among concurrent users in real time.

Full Stack Developer Aug 2024 – Jul 2025
MyWanderLust.in Delhi, India

- Delivered a Django/MySQL property management system and WhatsApp Business API notifications using Celery and RabbitMQ, contributing to a **35% increase in booking-driven revenue**.
- Improved reliability with structured logs, exception handling, idempotent retries, and Git-based workflows.

AI Data Engineer (Freelance) Sep 2024 – Feb 2025
Invisible AI Remote, India

- Trained and evaluated AI models on Telugu conversational data; partnered with reviewers to assess linguistic accuracy, cultural relevance, and safety, flagging quality issues before release.

Undergraduate Research Assistant Aug 2022 – Jan 2023
MIDAS Lab, IIIT Delhi New Delhi, India

- Accelerated Tamil dataset creation by **20%** with BLEU/agreement evaluation and morphosyntactic guidelines.
- Integrated scraping, CoNLL-U conversion, and Translate APIs into Python pipelines to ingest and validate Indic text.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Go, HTML/CSS
Frameworks: Django, Flask, React, Next.js, Bootstrap, Supabase
Data & Libraries: MySQL, MongoDB/Atlas (NoSQL), DynamoDB, Redis, Elasticsearch, Pandas, NumPy, NLTK, SciPy, Matplotlib, BeautifulSoup, Plotly Dash, Motor (async)
Messaging & APIs: Celery, RabbitMQ, SQS, REST, WhatsApp Business API, Nodemailer, tRPC, JSON, OAuth 2.0, SMTP
Cloud & DevOps: AWS (EC2, S3, Lambda, ElastiCache), Boto3, GCP Model Garden, Docker, Docker Compose, Git, GitHub Actions, cron, Apache Airflow
AI & Real Time: PyTorch, LangGraph, LangChain, OpenAI API, Llama, LoRA/SFT, vector search, LiveKit
Testing & Tools: Pytest, Jest, Playwright, Locust, Prometheus, Grafana, regex, multithreading, web scraping

Projects

Elastic Face Recognition on AWS Jan 2026 – May 2026
Python, Flask, PyTorch, Boto3, EC2, SQS, S3, DynamoDB

- Built a distributed inference service with S3 image storage, DynamoDB prediction caching, and paired SQS request/response queues; dynamically scaled **0–15 EC2 workers** with queue depth as demand rose and fell.

CityTraffic NLQ – Conversational Traffic Analytics Sep 2025 – Dec 2025
Python, Apache Airflow, MongoDB Atlas, Flask, React, Docker

- Owned infrastructure and database work for a team platform processing **2 million+ NYC collision records**, organizing geospatial indexes, replication, and geohash-based partitioning.
- Collaborated on daily Airflow ingestion through NYC Open Data's SODA/SoQL API and Flask endpoints supporting natural-language queries, vector search, and React/Leaflet.js/Plotly Dash visualizations.
- Evaluated spatial and temporal query performance under concurrent load using Locust, with Prometheus, Grafana, and Elasticsearch supporting platform monitoring and performance analysis.

Education

Arizona State University — M.S., Computer Science Aug 2025 – Present
IIIT Delhi — B.Tech., Electronics and Communication Engineering Aug 2020 – May 2024